Highlights

Artificial Intelligence for Customer Churn Prediction in Non-Contractual Retail Environments: A Systematic Literature Review

- Comprehensive SLR answering 8 research questions on AI applied to churn prediction
- Random Forest, XGBoost, and ANNs best methods; newer models remain under-explored
- Transactional, demographics, and behavioral features are the most commonly used
- Static definition of churn is most used, as opposed to customer-tailored ones
- Data scarcity, reproducibility, and model explainability still limit real-world use

Artificial Intelligence for Customer Churn Prediction in Non-Contractual Retail Environments: A Systematic Literature Review

Abstract

To retain customers effectively, retailers need accurate churn prediction to drive personalized engagement strategies. This study presents a systematic literature review of artificial intelligence applications for customer churn prediction in the retail sector, with a focus on non-contractual settings where churn must be detected behaviorally rather than explicitly recorded. From an initial pool of 1,370 articles identified across four major scientific databases, following a rigorous selection process, 41 relevant studies were selected and analyzed to answer 8 research questions drawn from our research goal. The review synthesizes findings on churn definitions, datasets, feature engineering practices, AI techniques, and evaluation metrics used for churn prediction. A notable trend is the reliance on static churn definitions based on fixed inactivity time windows, despite the large use of dynamic, personalized approaches that more accurately capture customer behavior. The widespread use of feature engineering methods, such as feature transformation, demonstrates the potential of this process to enhance model performance. Tree-based models and neural network architectures are identified as the most effective predictive techniques. However, heterogeneity in datasets, churn definitions, and modeling approaches poses challenges for performance comparisons. Additional issues include limited access to real-world datasets, class imbalance, and a lack of explainable AI tools. The research in this field is evolving to address these limitations with more sophisticated approaches. Future research directions proposed by authors include expanding model diversity (e.g., employing transformers, RNNs, and ensembles), improving explainability, and developing automated frameworks for feature selection and hyperparameter tuning.

Keywords: artificial intelligence, customer churn prediction, retail sector, non-contractual environment, systematic literature review

1. Introduction

~~In recent years, the retail sector has gone through significant transformations. This transformation~~, ~~primarily driven by the wide and fast adoption of digital technologies~~ [1]~~, has moved retail from the traditional physical stores to dynamic channel environments with ever-increasing focus on the online retail channel. While this evolution has brought opportunities, including broad customer reach, it has also introduced new challenges.~~

~~One such challenge is the lower barrier for customers to switch retailers. Historically, shopping was largely local, with small retailers serving specific neighborhoods, naturally limiting customer movement and making customer defection less of a concern. However,~~

~~the advent and rapid growth of e-commerce have made location-based convenience less relevant. Furthermore, e-commerce platforms have enabled consumers to effortlessly compare prices across retailers~~ [2, 3]~~, which has further intensified competition. The aforementioned barrier is further reduced by the the rise of large retailers and supermarkets, which have standardized product offerings~~ [4, 5]~~. As a result, a unique product selection is no longer a strong reason to prefer one option over another. Consequently, the likelihood of customer churn has increased significantly, making accurate churn prediction and proactive retention strategies indispensable.~~

Customer churn, namely the reduction or cessation of purchasing by existing customers, is a critical problem in non-contractual retail environments, where customers rarely signal defection explicitly. Instead, churn must be inferred from behavioural evidence such as declining purchase frequency, lower spending, or prolonged inactivity. This makes early churn detection both methodologically challenging and strategically important for retailers seeking to retain valuable customers.

~~The increased digitization of the retail sector~~ ~~, however, also provides a means to address the churn problem.~~ ~~Large volumes of~~ ~~customer interaction data and purchase histories are now stored digitally in retailers' data warehouses. This enables the training and deployment of AI and machine learning methods for the early detection of potential churners and for the development of targeting marketing intervention.~~ ~~In the early 2000s, customer churn prediction leveraging machine learning and AI has seen considerable application, initially in industries with contractual customer relationships, such as telecommunications~~ [6]~~.~~

~~The effectiveness of AI-driven churn prediction in contractual scenarios prompted its exploration in non-contractual retail environments, characterized by partial or gradual defection, where customers do not formally terminate relationships, but reduce their interactions or purchases over time~~ [7]~~. Predicting partial churn, particularly among behaviorally loyal customers, emerged as an important yet challenging goal, as these subtle shifts in consumer behavior often precede complete defection.~~

Churn prediction has become especially important in contemporary retail because digital transformation has lowered switching barriers, increased price transparency, and intensified competition across physical and digital channels. Customers can compare alternatives more easily and redistribute their purchases across retailers, while omnichannel journeys make disengagement more gradual and less immediately visible.

AI-based approaches are particularly relevant in this context because retailers now collect large volumes of transactional and interaction data that contain subtle signals of future defection. Rather than relying only on fixed heuristics, AI methods can learn complex, nonlinear, and temporal patterns from heterogeneous customer data. Examples include tree-based techniques such as Random Forest and XGBoost, as well as neural models such as Artificial Neural Networks and Long Short-Term Memory networks, all of which have been applied to churn prediction.

The retail sector is the connection between industries and consumers, accounting for 5% of GDP and employing 1 in 12 workers in the OECD member countries [8]. Considering both the macroeconomic weight and the economic implications, focusing on churn prediction in the retail sector is particularly relevant. Reichheld and Sasser show that even small improvements in customer retention can have substantial financial effects. In fact, with a 5% increase in retention rates, the profit can increase in the order of 25-95% across a wide range of business

contexts [9]. Against this background, an initial exploratory study conducted by our research group highlighted the relative scarcity of systematic literature reviews explicitly focused on the retail sector, motivating the present work, which, to the best of our knowledge, is the first systematic literature review (SLR) dedicated exclusively to AI-based churn prediction in non-contractual retail environments.

The goal of this SLR is therefore to identify, analyze, and synthesize the current state of research on the application of AI to customer churn prediction in the retail sector. In particular, our objective was to explore several dimensions of churn prediction, including definitions of churn, effective AI techniques, available datasets, feature engineering practices, performance metrics, challenges faced, and future research directions proposed by the surveyed studies. To achieve this goal, we designed and executed a rigorous research protocol inspired by the guidelines of Kitchenham and Charter [10]. The protocol includes research questions (RQs), a search string, a selection of relevant digital libraries, a set of inclusion and exclusion criteria, quality assessment criteria, and a data extraction form to collect evidence for answering the RQs. The protocol was executed on January 31, 2025, and, from an initial set of 1,370 articles, 41 primary studies were selected and analyzed.

The main contributions of this study are as follows:

- it provides a comprehensive review of the research on AI-based churn prediction methods in non-contractual retail environments;
- it highlights effective AI techniques and predictive features tailored to retail contexts;
- it discusses the main challenges faced, including data imbalance and model explainability;
- it identifies gaps and future research directions.

Ultimately, our review provides a valuable resource for both academics and practitioners, and offers guidance for future research and practical applications aimed at improving customer retention in this rapidly evolving research topic.

The rest of the paper is organized as follows: Section 2 provides essential background and defines key concepts related to customer churn and the AI techniques applied to churn prediction over time. Section 3 describes the research method in detail, including the protocol definition (Section 3.1) and its execution (Section 3.2). Section 4 presents the findings of our SLR, while Section 5 provides additional analysis and contextual reflections. Section 6 analyzes potential limitations and threats to validity. Finally, Section 7 concludes the paper by summarizing our main insights and proposing directions for future research.

2. Background

This section is divided into two parts. The first subsection outlines the fundamental concepts of customer churn, distinguishes between its various types, and traces the progression of churn management strategies. It highlights how the increasing availability of customer data and advancements in predictive analytics have transformed retention practices from more generalized retention marketing activities to more targeted approaches based on predicted risk of customer churn.

The second subsection presents an overview of the machine learning methods employed in the selected studies analyzed in this SLR, providing a reference point to help readers better contextualize and understand the subsequent sections of the document.

2.1. The customer churn problem and mitigation approaches

~~As stated by [7], "Churn analysis typically tries to define predictors of customer defection." Customer churn refers to the situation in which a customer terminates or substantially reduces their relationship with a firm, for example, by closing an account, switching to another provider, or no longer purchasing its goods or services. This behavior can occur in both contractual and non-contractual settings. In contractual industries, such as telecommunications or insurance, churn typically occurs when a customer cancels or fails to renew a formal service agreement, and is therefore observed as a form of total defection. In non-contractual contexts, customers can continuously adjust their purchase behavior without explicitly terminating the relationship. In these settings, churn is often studied through changes in purchasing patterns and partial defection, defines as a sustained reduction in purchase frequency or spending with the focal retailer that may eventually lead to complete defection.~~

As stated by [7], "Churn analysis typically tries to define predictors of customer defection." In the retail sector, customer churn refers to the substantial reduction or cessation of purchases from a focal retailer. Because retail is typically non-contractual, customers do not formally terminate the relationship. Instead, churn must be inferred from behavioural changes in purchasing activity. For this reason, churn in retail is often examined as partial or progressive defection, namely a sustained reduction in purchase frequency or spending that may eventually lead to complete defection.

Churn is commonly categorized into two main types: involuntary and voluntary [11]. Involuntary churn arises from factors outside the customer's control, such as relocation to an area not served by the business, financial hardship, or major life events like illness or death. Voluntary churn, on the other hand, occurs when a customer consciously decides to disengage. This may be driven by dissatisfaction with product or service quality, negative customer experiences, more competitive pricing or offerings from other providers, or a general perception of declining value.

While involuntary churn is often difficult for businesses to influence, minimizing voluntary churn is of strategic importance, particularly given that acquiring new customers is significantly more expensive than retaining existing ones [12]. Consequently, many businesses prioritize customer retention strategies and increasingly leverage churn prediction models to identify high-risk customers. One widespread approach was the introduction of loyalty programs, designed to raise the barrier to switching and encourage engagement [13]. These programs also gave retailers access to valuable data about customer behavior, enabling deeper insights into purchasing patterns [14].

With access to detailed customer data, businesses adopted more sophisticated strategies by leveraging historical customer behavior to segment individuals based on their predicted churn risk. This segmentation enabled the shift from broad marketing campaigns to more targeted efforts. Marketing budgets were increasingly directed toward customers identified as having a high risk of churn [15], allowing for the deployment of tailored interventions, such as personalized offers, loyalty incentives, or service enhancements to foster continued

engagement. In this context, accurate and early churn prediction emerged as a critical tool for optimizing resource allocation and improving customer retention outcomes.

Machine learning (ML) techniques have been applied to the task of churn prediction over the years [16]. ML involves the use of algorithms that learn patterns from data without explicit programming, allowing models to capture complex, often nonlinear relationships. The inherent complexity of churn prediction, coupled with the large volumes of data typically available in business contexts, makes this task particularly well-suited to ML approaches. ~~These techniques have been employed prevalently in contractual sectors including banking and telecommunications but less frequently in non-contractual environments such as retail.~~ Their application in retail is particularly challenging because churn must be inferred from customer behaviour rather than observed as an explicit termination event.

In the following section, we provide a structured overview of the AI and ML techniques employed across the selected studies, with a particular emphasis on their use in churn prediction within the retail context.

2.2. Machine learning techniques for churn prediction in the retail sector

~~Considering the large datasets available to retailers, ML is particularly well suited for churn prediction. Although our SLR considered primary studies employing any AI technique (e.g., deep learning, reinforcement learning), all studies that met our relevance and quality criteria ultimately applied ML methods.~~ Considering the large datasets available to retailers, ML is particularly well suited for churn prediction. The scope of this SLR was defined in terms of AI-based approaches and was not restricted a priori to machine learning. Nevertheless, after applying the search protocol, the inclusion and exclusion criteria reported in Section Appendix A.2, and the quality assessment criteria described in Section 3.1, the final set of included studies consisted exclusively of machine-learning-based approaches. This should therefore be interpreted as an empirical finding of the reviewed literature rather than as a methodological restriction imposed by the review design. ML, a subfield of AI, focuses on developing models capable of making predictions or decisions based on data-driven learning [17]. It encompasses a broad spectrum of techniques, each tailored to different problem domains and data characteristics. This section presents an overview and definitions of the machine learning methods identified in the selected studies included in this SLR, utilized for the task of customer churn prediction.

2.2.1. Tree-Based Methods

Tree-based methods utilize decision trees as their foundational structure, offering interpretability and the ability to handle both numerical and categorical data [18]. *Decision Trees* are non-parametric supervised learning algorithms used for classification and regression tasks. They work by recursively splitting the dataset into subsets based on feature values, forming a tree-like structure of decisions. *Random Forests* are ensemble methods that construct multiple decision trees during training and output the mode (for classification) or mean (for regression) of the predictions made by individual trees. *Extra Trees*, or Extremely Randomized Trees, introduce additional randomness by choosing split points randomly, which can lead to better generalization and reduced overfitting. *Boosting classifiers*, including *XGBoost*, *CatBoost*, *AdaBoost*, and *LogitBoost*, build strong learners by combining multiple weak learners, typically decision trees. *XGBoost* is known for its efficiency

and performance, particularly on structured data, while *CatBoost* handles categorical variables natively, reducing preprocessing efforts. *AdaBoost* improves classification by adjusting weights to emphasize previously misclassified instances. *LogitBoost* optimizes a logistic loss function through additive modeling. *Bagging classifiers*, such as Bootstrap Aggregating, train multiple models on different bootstrap samples of the data and combine their outputs to improve stability and accuracy. Tree based methods demonstrate strong performance when trained on an imbalanced dataset. For the churn prediction problem, where the ratio of churners is significantly lower than that of non-churners, this characteristic of tree-based methods would be advantageous [19, 20].

2.2.2. Distance-Based Methods

Distance-based methods rely on geometric interpretations in feature space [21]. *Support Vector Machines (SVMs)* construct optimal hyperplanes to separate classes with maximum margin and are effective in high-dimensional spaces. *K-Nearest Neighbors (KNN)* classifies data points based on the majority class among the 'k' closest examples in the training set, offering simplicity and interpretability. *Multivariate Adaptive Regression Splines (MARS)* [22] is a flexible regression technique that captures nonlinear relationships by fitting piecewise linear regressions, making it suitable for modeling complex interactions between variables such as with churn prediction.

2.2.3. Linear and Discriminant Analysis

The methods in this category are often classification and regression tasks. *Ridge Regression* is a technique that adds L2 regularization to linear regression, penalizing large coefficients to prevent overfitting. Linear Discriminant Analysis and Quadratic Discriminant Analysis are both derive a discriminant function for each class and classify observations based on the function's output [23]. *Linear Discriminant Analysis (LDA)* is a method that finds a linear combination of features that best separates two or more classes. *Quadratic Discriminant Analysis (QDA)* is similar to LDA but allows for each class to have its own covariance matrix, capturing more complex class distributions. *Logistic Regression* is a statistical model that predicts the probability of a binary outcome using a logistic function [24]. *Boosted Logistic Regression* enhances logistic regression by applying boosting techniques to improve predictive performance. These methods are perhaps simplistic for the churn prediction problem at hand, due to their limited ability to generalize relationships.

2.2.4. Stochastic

Stochastic methods are probabilistic models that make predictions based on the likelihood of outcomes, incorporating randomness in their processes. *Naive Bayes* [25] includes classification techniques based on Bayes' Theorem, assuming independence between predictors. *Poisson Processes* [26] are stochastic processes that model the occurrence of events over time, often used for count data. Although such methods can be effective for the right task, the assumptions made in these probabilistic models might not apply to the churn prediction problem, where there can be complex and dependent interrelations between the feature variables [27].

2.2.5. Neural Networks

Neural networks are inspired by the structure and functioning of the human brain and consist of layers of interconnected nodes (neurons) that process input data [28]. *Artificial Neural Networks (ANNs)* are composed of multiple layers of neurons and are used for a wide range of prediction tasks. *Recurrent Neural Networks (RNNs)* are designed to process sequential data by incorporating recurrent connections, where the output from one time step is fed as input to the next. This structure enables RNNs to capture temporal dependencies in data such as time series or text. *Long Short-Term Memory (LSTM)* networks, a specialized form of RNN, address the limitations of standard RNNs by incorporating memory cells and gating mechanisms, allowing them to learn long-term dependencies more effectively. *Convolutional Neural Networks (CNNs)*, typically used in image processing tasks, consist of convolutional layers that automatically learn spatial hierarchies of features. *Restricted Boltzmann Machines (RBMs)* are generative stochastic neural networks that learn probability distributions over input data. They are particularly useful for dimensionality reduction, feature learning, and collaborative filtering. Neural networks can have great generalization power due to their ability to model complex relationships, however, this generalizing ability comes at the cost of requiring a large corpus of data to extract the best performance form such models. For the churn prediction problem, retailers naturally have access to data at this scale; therefore, neural network-based approaches could be well suited [29].

3. Materials and method

This study follows a research protocol based on the SLR guidelines of Kitchenham and Charters [10] and complies with the PRISMA 2020 reporting standard [30], thus ensuring a structured, transparent, and reproducible review process. In Section 3.1, we present the details of the protocol, including: objectives, research questions, search strategy (digital libraries and search strings), inclusion and exclusion criteria, quality assessment metrics, and data extraction form. Subsequently, Section 3.2 reports the results obtained from applying the protocol.

3.1. Protocol definition

3.1.1. Goal and research questions definition

The goal of this SLR is *to explore the state of the art regarding the application of AI technologies in customer churn prediction in the retail sector, focusing on the techniques employed, the data used, and the perceived effectiveness and limitations.*

To achieve this goal, we formulated 8 Research Qestions (RQs). These RQs directly address the core research objectives of our SLR by analyzing the contributions, methodologies, and findings of the selected primary studies. The 8 RQs are defined as follows.

RQ1: What are the adopted definitions of customer churn?

RQ2: What features have been used for customer churn prediction, and which have been found to be more effective?

RQ3: How have the available features been pre-processed and/or engineered before analyzing them using AI techniques?

RQ4: What AI techniques have been applied and found to be most effective in predicting customer churn in the retail sector?

RQ5: What are the most commonly used metrics to evaluate the performance of churn prediction techniques?

RQ6: Which publicly available datasets have been used for retail customer churn prediction, and what are their main characteristics?

RQ7: What are the challenges and limitations reported in using AI techniques for churn prediction?

RQ8: What are the emerging trends and future directions for research in this area?

3.1.2. Digital libraries selection

To identify eligible studies, we queried four premier digital libraries commonly employed in systematic literature reviews: Scopus, Web of Science (WoS), ACM, and IEEE Xplore. Scopus (Elsevier) was chosen for its expansive, interdisciplinary coverage (encompassing over 300+ subject areas) and its stringent selection of peer-reviewed works[1]. The WoS Core Collection (Clarivate) comprises more than 22,000 journals, applies rigorous evaluation criteria, and has only about 10–12% acceptance rate for the Science Citation Index Expanded[2]. ACM and IEEE are leading professional organizations that publish key journals and conference proceedings, and organize major international scientific conferences in computer science and on artificial intelligence in particular.

3.1.3. Search string formulation

To systematically construct the search string for identifying primary studies relevant to our SLR, we followed the PICOC (Population, Intervention, Comparison, Outcome, Context) framework proposed by Petersen et al. [31]. In our case, the *Population* is represented by *Customers* of the retail sector. The *Intervention* focuses on *Artificial Intelligence* techniques, and the desired *Outcome* is the *Churn Prediction*. The *Comparison* viewpoint does not apply to our study as we are not interested in comparing different interventions (e.g., traditional and AI-based approaches to customer churn prediction) while the *Context* viewpoint is represented by the *Retail Sector*, as we are interested in surveying the literature on applications of AI to predict churn of retail customers.

For each main term within the PICOC viewpoints, a list of relevant synonyms, abbreviations, and alternative spellings was identified. The search string was then constructed by combining these terms using AND and OR operators, customized to match the syntax required by the search engine of each selected digital library, and applied to the title, abstract, and keywords fields. Following Kitchenham et al.'s [10] guidelines, the search string was also validated using three control papers [32, 33, 34] manually selected during a preliminary pilot study.

The resulting PICOC and the search strings customized for the different digital libraries are reported in Appendix A.1.

3.1.4. Inclusion and exclusion criteria definition

Inclusion and exclusion criteria (IC/EC) were defined to delineate the scope of the review. A paper retrieved from the selected digital libraries using the defined search string will be

[1] https://www.elsevier.com/products/scopus/content#3-selection-standards

[2] https://clarivate.libguides.com/librarianresources/coverage

retained only if it satisfies all IC and does not meet any EC [10]. Specifically, studies will be excluded if they are position papers or short papers; if they are not peer-reviewed; if the full text is not accessible; if they are secondary or tertiary studies such as reviews, mini-reviews, or surveys; if they are duplicates or earlier versions of another candidate paper; or if they primarily replicate or test the methodology of other authors rather than presenting original research.

To qualify for inclusion, a publication must be written in English, focus specifically on customer churn prediction using artificial intelligence methods, utilize datasets from the retail sector, and base its predictions on customer and/or transactional purchase data.

The IC/EC criteria are reported in detail in Appendix A.2.

3.1.5. Quality assessment criteria definition

To ensure that the studies included in this SLR provide methodologically sound and reliable evidence, we defined and applied a set of six Quality Criteria (QCs) following the guidelines proposed by Kitchenham et al. [35]. These criteria assess the methodological rigor, clarity of reporting, and validity of the findings, supporting a transparent evaluation that strengthens the credibility of the review's conclusions. The list of defined QCs, along with the possible scores and their associated rationales, is presented in Table 1. The overall quality of each study was determined by summing these scores.

Table 1: Quality assessment criteria

	Possible scores		
Quality Criteria	**Fully satisfied (+1.0)**	**Partially satisfied (+0.5)**	**Not satisfied (+0.0)**
QC1: Is the paper published in a qualified venue?	The paper is published in a journal ranked in Q1 or Q2 quartile or in a CORE A+ or A Conference	The paper is published in a venue ranked in Q3 quartile or in a CORE B Conference	The paper is published in a venue ranked in Q4 quartile or in a CORE C or D Conference
QC2: Does the work clearly describe which specific AI techniques have been used to predict Customer Churn?	The adopted AI technologies are described in detail (not only the specific algorithms are specified, but also details about hyperparameters and experimental settings are provided)	The adopted AI techniques are clearly specified but in a general way, i.e., no additional information on how they were used is provided (such as, hyperparameters setting, experimental setting, etc.)	No specific AI techniques are mentioned (e.g., only the use of clustering or regression methods are mentioned)
QC3: Does the work clearly describe which dataset/variables are used to predict Customer Churn?	The dataset used for churn prediction is available for download	The dataset used for churn prediction is described in detail but is not available for download	The dataset used for churn prediction is only summarily described or not described at all
QC4: Is class imbalance dataset addressed?	Class imbalance is handled with appropriate methods and its impact is clearly discussed	Class imbalance is mentioned and partially addressed, but with limited detail or discussion	Class imbalance is not mentioned or addressed
QC5: Have the adopted AI techniques been experimentally evaluated?	The performance of the adopted AI techniques has been evaluated using a rich set of appropriate metrics	The performance of the adopted AI techniques has been evaluated using a few metrics	The performance of the adopted AI has not been evaluated at all
QC6: Are the results of the study discussed?	The results are discussed in detail, with clear summaries provided	The results are mentioned but not thoroughly discussed or interpreted.	The results are not discussed or are only mentioned without meaningful analysis.

3.1.6. Data extraction form

To facilitate the extraction of evidence from the selected sources, we designed the Data Extraction Form (DEF) reported in Table 2. This form specifies the key pieces of evidence to be identified in the selected papers and systematically recorded to answer each of the defined RQs. To support collaboration among the authors, the DEF was implemented as an online Google spreadsheet.

Table 2: Data extraction form.

RQs	Evidence to be extracted
RQ1	List of sentences describing the adopted definitions of "Customer Churn"
RQ2	List of sentences identifying data/variables/features and the most effective features used for Churn Prediction
RQ3	List of sentences identifying data aggregation or correlation, or combination methods (e.g., based on time periods, order total value, products' categories, seasonality, customer segmentation, etc.) that have been used before applying AI techniques on the resulting features
RQ4	List of sentences identifying the AI techniques used and found to be most effective for Churn Prediction in the Retail domain
RQ5	List of sentences identifying the metrics used to evaluate the performance of the adopted AI techniques.
RQ6	All the information related to the dataset used for Churn Prediction and publicly available (publication date, URL, # of records, # of customers, fields included, etc.)
RQ7	List of sentences on identified and discussed challenges and limitations
RQ8	List of sentences on emerging trends and future research directions

3.2. Protocol execution

This section provides an overview of the study conduction, following the protocol designed in Section 3.1. Specifically, we first provide details about the selection process and quality assessment of the included papers, followed by details on the data extraction process.

3.2.1. Studies Selection Process

The process followed to select primary studies is illustrated in Figure 1. The Figure depicts the sequence of executed steps and their outcomes in terms of retrieved, discarded, and finally selected papers.

1. *Querying of digital libraries*: In this initial step, the search strings (introduced in Section 3.1) were used to query the four digital libraries selected for our SLR (ACM, IEEE Xplore, Scopus, and WoS). No restrictions were applied on the publication year during the search, which was conducted on January 31st, 2025, resulting in a total of $1,370$ records.

2. *Removal of duplicate records*: From the initial $1,370$ retrieved records, **316** duplicate entries were identified and removed, leaving $1,054$ studies for screening.

3. *Application of Inclusion and Exclusion Criteria (IC/EC) — first step*: The remaining 1,054 records were split into two groups, and their titles, abstracts, and metadata were independently screened by two researchers applying all IC and EC defined, except for EC6 and IC2, which require full-text analysis. During this phase, 912 studies were excluded by mutual agreement, resulting in 142 studies progressing to the next step.

4. *Application of IC and EC on full texts*: The 142 papers were divided between the two researchers, who performed a full-text assessment applying all the IC/EC including EC6 and IC2. This step resulted in the exclusion of 66 studies and the retention of 76 papers, of which 16 were classified as "doubtful" due to disagreement between reviewers.

5. *Resolution of doubtful studies*: To resolve the doubtful cases, two additional researchers joined the initial reviewers for a joint discussion and re-assessment based on full-text analysis. After this collaborative review, 14 studies were excluded, and 2 were included, resulting in a final set of 62 selected studies.

3.2.2. Quality assessment

Each of the 62 selected papers was independently evaluated by two researchers using the six QCs described in Section 3.1. Papers that received a total score below 3 from both researchers were excluded, while those with substantially differing scores or borderline cases (scores between 3 and 4) were discussed collectively by all researchers. Following this process, 41 studies were retained as the final set of high-quality primary studies included in this review. The complete list of selected papers along with their quality scores is presented in Table 3.

3.2.3. Data extraction

Given that our RQs (see Section 3.1) span the distinct domains of AI and marketing, the review team was composed of experts in both fields. The team collaborated to extract relevant data from each of the 41 selected studies, utilizing the data extraction form presented in Table 2. In cases of uncertainty, disagreements were resolved through discussion to reach a consensus.

The data extraction form and the collected data are publicly accessible online.[3]

[3] https://docs.google.com/spreadsheets/d/1d6RD3cnYkc6AYOfw-6SbhhxwG-dBsKlVmFowj5EHDg8/edit?usp=sharing

Table 3: Quality criteria assessment and resulting selected studies.

		Quality Criteria (QC)						
ID	**Study Title**	**1**	**2**	**3**	**4**	**5**	**6**	**Total Score**
S1	Customer data prediction and analysis in e-commerce using machine learning [36]	0.5	1	0.5	0	0.5	1	**3.5**
S2	E-commerce big data processing based on an improved RBF model [37]	1	0.5	0.5	0	1	1	**4**
S3	Comparative SHAP Analysis on SVM and K-NN: Impacts of Hyperparameter Tuning on Model Explainability [38]	0	1	1	0.5	1	1	**4.5**
S4	Analyzing Customer Churn Behavior Using Data mining Approach: Hybrid Support Vector Machine and Logistic Regression in Retail Chain [39]	0.5	1	0.5	0	1	1	**4**
S5	TSUNAMI - an explainable PPM approach for customer churn prediction in evolving retail data environments [40]	0.5	0.5	1	1	1	1	**5**
S6	Predicting customer churn in the fast-moving consumer goods segment of the retail industry using deep learning [41]	0	1	0.5	0.5	0.5	1	**3.5**
S7	Explainable machine learning models applied to predicting customer churn for e-commerce [42]	0.5	0.5	1	0	0.5	0.5	**3**
S8	Prediction of Customer Behavior Changing via a Hybrid Approach	1	0.5	1	0.5	1	1	**5**
S9	Vision Transformer Approach to Customer Churn Prediction Radar Chart Image Classification for Non-subscription Based E-commerce [43]	0	1	0.5	1	0.5	1	**4**
S10	Development of fading channel patch based convolutional neural network models for customer churn prediction [44]	0.5	1	1	0.5	1	1	**5**
S11	Deciphering Digital Social Dynamics: A Comparative Study of Logistic Regression and Random Forest in Predicting E-Commerce Customer Behavior [34]	0.5	1	0.5	0.5	0.5	1	**4**
S12	A Predictive Model for E-Commerce Customer Churn under an Intelligent Algorithm [45]	0	0.5	0.5	1	0.5	0.5	**3**
S13	E-commerce customer churn prevention using machine learning-based business intelligence strategy [46]	0.5	0.5	0.5	1	0.5	0.5	**3.5**
S14	Customer Churn Prediction Using Ordinary Artificial Neural Network and Convolutional Neural Network Algorithms: A Comparative Performance Assessment [47]	0.5	1	0.5	0	1	0.5	**3.5**
S15	Novel time slicing approach for customer defection models in e-commerce: a case study [48]	1	0.5	0.5	0.5	0.5	0.5	**3.5**
S16	Customer Churn in Retail E-Commerce Business: Spatial and Machine Learning Approach [33]	1	1	1	0.5	1	1	**5.5**
S17	XGBoost-Based E-Commerce Customer Loss Prediction [49]	1	1	0.5	1	1	1	**5.5**
S18	An Improved Machine Learning Based Customer Churn Prediction for Insight and Recommendation in E-commerce [50]	0	0.5	0.5	0.5	0.5	1	**3**
S19	Churn Detection Using Machine Learning in the Retail Industry [32]	0	1	0.5	1	1	1	**4.5**
S20	Performance Evaluation of Various Classification Techniques for Customer Churn Prediction in E-commerce [51]	0.5	1	1	0.5	1	1	**5**
S21	Customer Churn Prediction Using Deep Learning [52]	0	1	0.5	0	1	0.5	**3**
S22	A PCA-AdaBoost model for E-commerce customer churn prediction [53]	1	0.5	0.5	0.5	1	1	**4.5**
S23	Deep learning for customer churn prediction in e-commerce decision support [54]	0.5	1	0.5	0.5	1	0.5	**4**
S24	A Clustering-Prediction Pipeline for Customer Churn Analysis [55]	0.5	0.5	1	1	0.5	1	**4.5**
S25	A New Time Series Approach in Churn Prediction with Discriminatory Intervals [56]	0	0.5	1	0.5	1	1	**4**
S26	User Churn Model in E-Commerce Retail [57]	0	1	1	0	1	1	**4**
S27	Customer Churn Prediction in FMCG Sector Using Machine Learning Applications [58]	0	0.5	0.5	1	0.5	0.5	**3**
S28	Predicting Customer Turnover Using Recursive Neural Networks [59]	1	0.5	1	0	1	1	**4.5**
S29	User Modeling for Churn Prediction in E-Commerce [60]	1	0.5	0.5	0.5	0.5	0.5	**3.5**
S30	Clustering prediction techniques in defining and predicting customers defection: The case of e-commerce context [61]	1	0.5	0.5	0.5	0.5	1	**4**
S31	Predicting partial customer churn using Markov for discrimination for modeling first purchase sequences [62]	1	0.5	0.5	0.5	0.5	0.5	**3.5**
S32	Research on e-commerce customer churning modeling and prediction [63]	0	1	0.5	0.5	0.5	0.5	**3**
S33	Modeling partial customer churn: On the value of first product-category purchase sequences [64]	1	0.5	0.5	0	0.5	0.5	**3**
S34	An extended support vector machine forecasting framework for customer churn in e-commerce [65]	1	0.5	0	1	0.5	0.5	**3.5**
S35	Customer attrition in retailing: An application of Multivariate Adaptive Regression Splines [66]	1	0.5	0.5	0	0.5	1	**3.5**
S36	Applying data classification techniques for churn prediction in retailing [67]	0.5	0.5	0.5	0	0.5	1	**3**
S37	Customer base analysis: Partial defection of behaviourally loyal clients in a non-contractual FMCG retail setting [7]	1	0.5	0.5	0	1	1	**4**
S38	Handling class imbalance in customer churn prediction	1	0.5	0	1	1	1	**4.5**
S39	A Customer Churn Prediction Model for Hartlief Shop & Bistro, Namibia [68]	0	0.5	0.5	0.5	1	1	**3.5**
S40	Comparative Study of Customer Churn Prediction Based on Data Ensemble Approach [69]	0	0.5	0	0.5	1	1	**3**
S41	Predicting Customer Churn in Retailing [70]	0.5	0.5	0.5	0	1	1	**3.5**

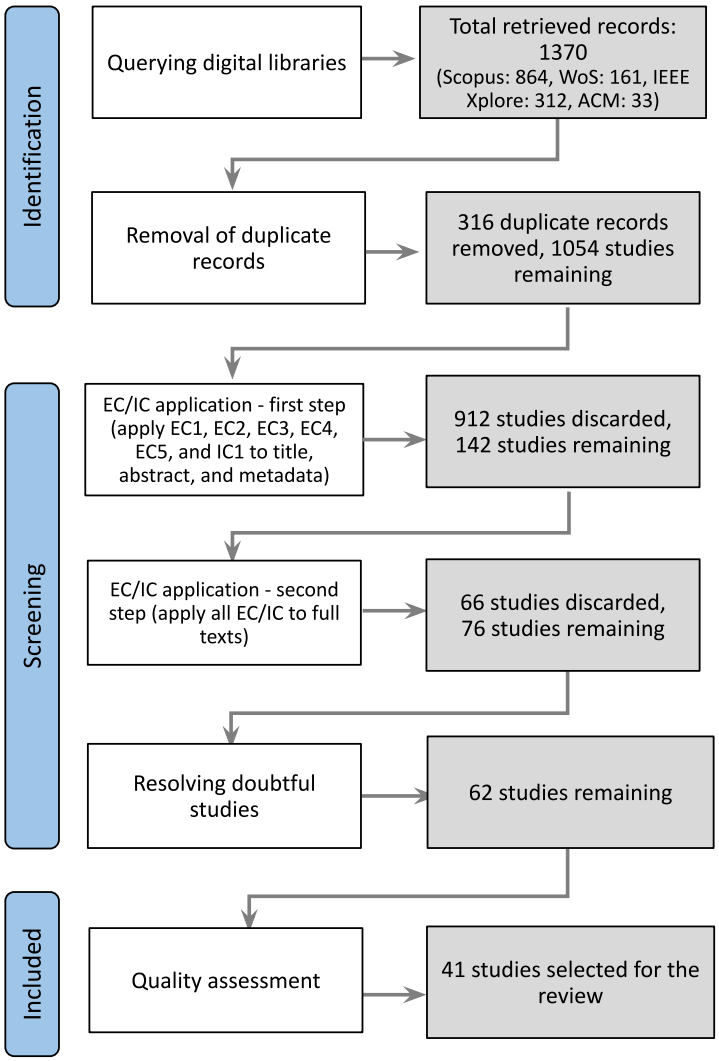

Figure 1: Studies selection process.

4. Results

This section presents the results of our study, derived from analyzing the evidence extracted from the selected 41 studies and answering the defined RQs. It is worth noting that some interpretations may assume that the trends observed among the selected papers are indicative of broader publication patterns in the field.

4.1. Descriptive analysis of included studies

We present here the trends pertaining to the publication space of the selected studies. The information regarding the venue name, venue type, publication year, and research area is tabulated in Table 4. Despite not applying any restrictions on the publication date during the selection process, the earliest paper included in the analysis was published in 2005. Of the 41 selected papers, 31 (76%) were published from 2021 onward, which indicates a growing interest in this topic in recent years. It is important to note that the low number of papers

for 2025 is due to the fact that the paper extraction was carried out at the end of January; therefore, only papers published before this date were included. Among the 41 selected papers, 28 were published in journals, and 13 in conference proceedings. The higher number of journal publications in the selected studies may be due to a higher likelihood of meeting the quality standards employed in this study. With the exception of four publications in the journal "Expert Systems with Application", the remaining 37 studies were each published in a distinct venue. In regards to the research areas of the publication venues, as expected for research on customer churn prediction using artificial intelligence, the dominant subject area associated with the publication venues of the selected studies was Computer Science with 29 studies, followed by Engineering with 19 studies, and subsequently by "Mathematics, Physics, and Astronomy" with 11 studies. "Business, Economics, and Finance" appeared in 7 studies, while Medicine and Neuroscience, Material Sciences, and the Multidisciplinary category were the least represented fields, with only 3, 2, and 1 associated studies, respectively. It is worth noting that each venue can be associated with multiple subject areas, meaning that the total number of subject-area associations exceeds the 41 selected studies. Since not all conferences provide publicly available information on their subject areas, therefore, a procedure was adopted to ensure consistency in the analysis. The areas of interest for journal articles were obtained from Scimago[4]. For conferences lacking this information, the corresponding subject areas were retrieved from Scopus[5].

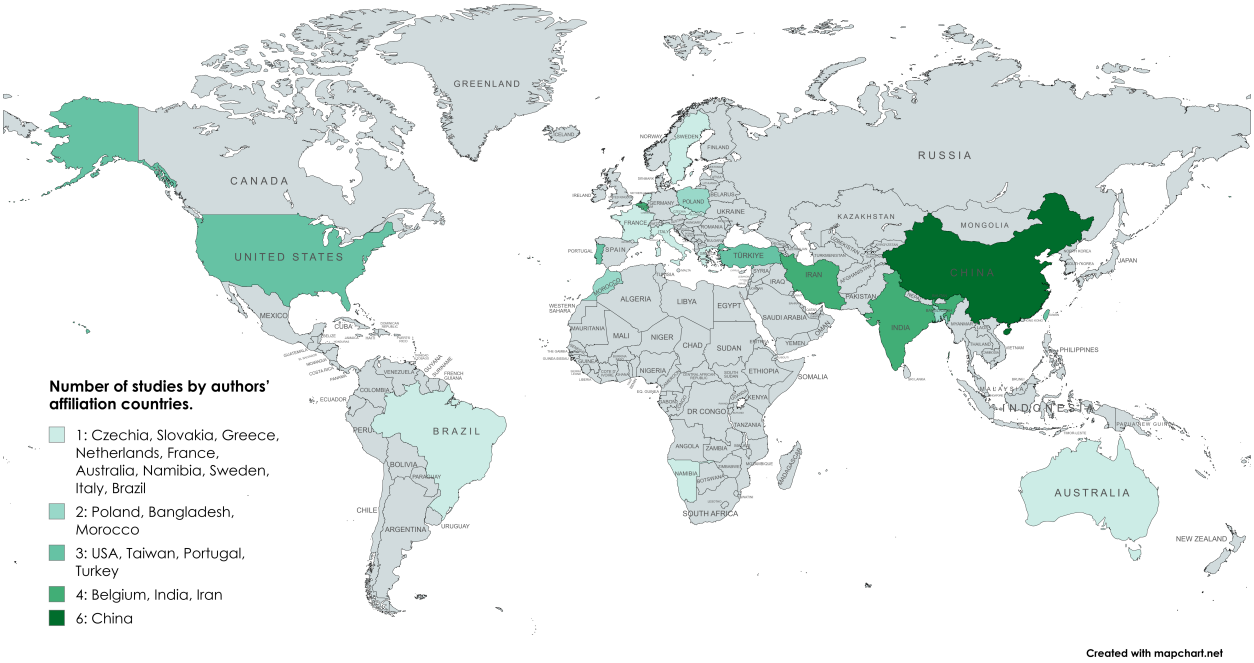

Figure 2: Distribution of selected studies by country of affiliation of authors.

Figure 2 presents the results of the author's affiliated country. Each country is counted only once per study, regardless of the number of authors affiliated with that country. If a

[4] https://www.scimagojr.com
[5] https://www.scopus.com

study has authors from multiple countries, it is assigned to each of those countries. China had the highest representation, with six associated publications, followed by India, Belgium, and Iran, each contributing four publications. The United States, Taiwan, Portugal, and Turkey, each with three associated publications. Bangladesh, Morocco, and Poland contributed two publications each, while the remaining 10 countries were represented by a single publication. It is worth noting that most publications originate from Asia, highlighting the region's particularly strong interest in the topic.

Table 4: Bibliometrics of the included studies

Study IDs	Year	Venue Name	Computer Science	Engineering	Mathematics, Physics and Astronomy	Decision Sciences	Business, Economics and Finance	Medicine and Neuroscience	Materials Science	Multidisiplinary
Conferences (13 Studies)										
S9	2025	Lecture Notes in Computer Science	●		●					
S39	2024	International Conference on Emerging Trends in Networks and Computer Communications		●						
S40	2023	Intelligent Computing and Control for Engineering and Business Systems	●	●	●	●		●		
S3		International Conference on Information Integration and Web Intelligence	●							
S18	2022	International Conference on Computer and Information Technology	●	●		●		●		
S19		2nd International Conference on Computer, Control and Robotics	●	●	●					
S41		International Conference on Machine Learning and Applications	●							
S21	2021	Advances in Intelligent Systems and Computing	●	●						
S23		International Conference on Business Information Systems				●	●			
S24		International Conference on Knowledge Science, Engineering and Management	●		●					
S25		International Conference on Computer Engineering and Knowledge	●							
S27		International Workshop on Artificial Intelligence for Knowledge Management	●			●				
S36	2007	International Conference on Electronic Business	●				●			
Journals (28 Studies)										
S7	2025	International Journal of Artificial Intelligence	●	●		●				
S1	2024	Bulletin of Electrical Engineering and Informatics	●	●	●					
S2		Journal of Intelligent Systems	●							
S4		International Journal of Research in Industrial Engineering		●		●				
S5		Journal of Intelligent Information Systems	●							
S6		Mathematics and Computational Sciences			●					
S10		International Journal of Systems Assurance Engineering and Management		●			●			
S11		Journal of Applied Data Sciences	●							
S8	2023	IEEE Open Journal of the Computer Society	●							
S12		UPB Scientific Bulletin			●					
S13		Measurement: Sensors		●					●	
S14		Gazi University Journal of Science		●						●
S15	2022	Data Science and Management	●			●	●			
S16		Journal of Theoretical and Applied Electronic Commerce Research	●				●			
S17		Computational Intelligence and Neuroscience	●		●			●		
S20		Microprocessors and Microsystems	●							
S22		Annals of Operations Research							●	
S26		Scientific Papers of the University of Pardubice, Series D					●			
S28	2021	Wireless Communications and Mobile Computing	●	●						
S29	2019	IEEE Intelligent Systems	●							
S30	2018	International Journal of Electrical and Computer Engineering		●		●	●			
S32	2014	Open Cybernetics and Systemics Journal		●	●					
S35	2013	Expert Systems with Applications	●	●						
S31	2012	Advances in Data Analysis and Classification	●		●					
S33		Expert Systems with Applications	●	●						
S34	2011	Expert Systems with Applications	●	●						
S38	2009	Expert Systems with Applications	●	●						
S37	2005	European Journal of Operational Research	●	●	●	●				
			29	**19**	**11**	**9**	**7**	**3**	**2**	**1**

4.2. Answers to Research Questions (RQs)

4.2.1. RQ1: What are the adopted definitions of customer churn?

In non-contractual settings such as Fast Moving Consumer Goods (FMCG) and e-commerce, defining churn presents a significant challenge. Unlike contractual industries, where customers formally terminate their relationship, churn in non-contractual sectors is a behavioral inference rather than a clearly observable event. Our review found that more than one-third of the 41 surveyed studies did not provide a clear definition of churn. In many cases, this omission was due to the use of pre-processed or third-party datasets, particularly those sourced from platforms such as Kaggle[6], where churn is provided as a pre-labeled flag with minimal or no documentation or behavioral context. Moreover, several studies relied on external datasets or cited sources that were either inaccessible or insufficiently described, rendering their churn definitions non-replicable. Even when a definition was provided, it was often vague or overly simplistic, lacking the detail necessary for methodological transparency and reproducibility.

Table 5: Adopted churn definitions and associated studies.

Churn definition	**Study ID(s)**
Static	S3, S5, S10, S13, S14, S15, S20, S21, S22, S25, S27, S39, S41
Dynamic	S6, S19, S30, S31, S33, S35, S36, S37
Static with dynamic window	S16, S23
Dynamic with dynamic window	S24
Cluster/Segment based	S9
Probabilistic based	S8

To classify heterogeneous definitions in a consistent way, we conducted a thematic analysis of the churn definitions reported in the selected studies. To the best of our knowledge, no systematic typology of churn in non-contractual retail contexts exists. Therefore, our analysis focuses on the operational definitions of churn used in practice rather than on new conceptual types of churn. We used the data extraction form described in Section 3 to collect, for each study, the textual definition of churn and the details of its specifications. We then iteratively coded each study along two main dimensions: *i.* how the temporal window used to assess churn is defined (fixed length that is identical for all customers vs. implicitly or explicitly customer-specific), and *ii.* how churn is determined within that window (fixed heuristic thresholds on inactivity or spending, behaviour-based thresholds relative to a reference period, or data-driven mechanisms such as clustering or probabilistic models). By systematically combining the values of these two dimensions and applying the resulting scheme to all 26 studies that provide an explicit operational definition of churn, we derived a data-driven taxonomy of six archetypal churn definitions. This taxonomy is summarized in Table 5, together with the associated Study IDs, and provides a unifying lens to map heterogeneous definitions found in the literature. Although it is grounded in our sample, the taxonomy is general enough to be used as a reference framework in future research on churn prediction in non-contractual retail environments. The following categories of churn definitions were identified:

[6] kaggle.com

- *Static definition*: This is the most common and straightforward approach. Churn is defined based on a fixed window period (e.g., 90 days, 3 months, or 1 year) during which the absence or reduction of customer activity is interpreted as churn. The threshold (e.g., no spending or a significant drop in purchases) is applied uniformly across all customers, regardless of their behavioral patterns. While this method is easy to implement and interpret, it lacks personalization and may overlook individual variations in engagement.

- *Dynamic definition*: This method employs fixed-length observation and labeling windows (e.g., P1 = 3 months, P2 = 3 months). Churn is identified based on significant changes in customer behavior between these two periods. For example, a customer is labeled as churned if their loyalty in P2 declines sharply compared to their behavior during P1, or if their spending in P2 falls below a threshold that is behaviorally derived from P1. This approach captures behavioral shifts and partial churn more effectively than static definitions.

- *Static definition with dynamic window*: This approach evaluates churn based on an observation window whose start and/or length are dynamically determined by an event specific to the customer's purchase history (e.g., the customer's first or most recent transaction), while the threshold used to label churn (e.g., no purchases) remains static and uniformly applied to all customers. This approach introduces some adaptability in identifying the observation period while preserving the simplicity of a static threshold.

- *Dynamic definition with dynamic window*: This is the most flexible and personalized approach. Both the observation window and the churn threshold are determined based on each customer's unique behavioral history. For example, the window start point can be determined by their second last purchase and the dynamic recency threshold for churn can be determined for each customer as a multiple of their last two transactions.. This fully dynamic strategy enables highly tailored churn definitions and is commonly adopted in advanced models that prioritize precision over simplicity.

- *Cluster/segment-based*: In this category, churn is inferred through data-driven segmentation techniques, such as K-means clustering, which group customers based on behavioral patterns. Customers assigned to groups exhibiting a complete absence or substantial reduction of activity are labeled as churners. This unsupervised approach does not rely on predefined thresholds, but rather identifies churn through emergent patterns in the data. Only one study employed this definition.

- *Probabilistic-based*: This approach uses statistical or machine learning models to estimate the probability that a customer remains active. When this probability falls below a predefined threshold, the customer is classified as a churner. This method allows for dynamic adjustment based on differing consumption patterns and cycles. Among the reviewed studies, only one adopted this method.

It is interesting to note that this simple approach dominates the literature with 13 studies using this method. This is followed by the dynamic definition method, which is utilized by 9 studies. The static definition with dynamic window approach is used by 2 studies. The other

more bespoke approaches, including probabilistic and segmentation methods, are utilized only by one study.

4.2.2. RQ2: What features have been used for customer churn prediction, and which have been found to be more effective?

The analysis of the 41 selected studies revealed a wide range of features employed for customer churn prediction, with some proving particularly effective in capturing churn-related behavior. Due to inconsistent terminology used across studies, feature names were standardized where possible to enable consistent comparison. When only intermediary or derived variables were reported, they were traced back and mapped to their most likely source features to align with a common feature taxonomy.

Variables such as *Customer ID*, *Transaction ID*, and other similar identifiers were excluded, as they do not provide meaningful information for churn prediction. These variables are primarily used within the data processing pipeline to consolidate records at the customer level.

The top section of Table 6 lists the features used for churn prediction across the selected studies, indicating for each study whether they were used and, if applicable, whether they were identified as top features. Empty circles (○) denote intermediary or derived variables, while filled circles (●) mark features explicitly identified as most important in the studies. Columns corresponding to studies with unclear source features are shaded in light grey.

The bottom section summarizes the use of feature categories. Each category is marked for a study if at least one variable from that category was included in the study. For studies in which the source features could not be fully identified but enough details were available to determine usage at the category level, only the relevant categories are indicated. These columns are also shaded in grey.

Table 6: Features used for churn prediction, with top-ranked features highlighted.

Feature Categories	Features	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	S17	S18	S19	S20	S21	S22	S23	S24	S25	S26	S27	S28	S29	S30	S31	S32	S33	S34	S35	S36	S37	S38	S39	S40	S41
Geo Demographic	Location															●		○									●	○						○							○
	User Location - Population															●			○																						
	User Location - City Tier			●			○				○																														
	User Location - Rural or Urban															○																									
	User Location - % Immigrants															○																									
	User Location - Population Earning Dist																		○																						
Demographic	Gender	○		○	○		○				○	○						○	○								●					●		○	○						○
	Age				○							○		○				○	○								○	○				●		○	○						
	Location to Home Distance			○			○				●																														
	Resident Status													○																											
	Marital Status	○		●	●		●				○							○									○					○									
	Education Information				○									○													○					○									
	Employment Information				○									○													○							○								
	Childern Information													○																					○		○					
	Family Member Information				○									○																					○		○					
	Posession of Devices													○																												
	Credit Score																															●										
	Buying Power				●							○						●														○										
	Number of Addresses	○		○			○				○							○																								
	Number of Registered Devices			○			○				○							○																								
	Way of Introduction to Organization				●																																					
	Membership Level											○																														
Behavioral	Active																	○																								
	Purchase Channel (TS)				●							○																													○	
	Payment Method (TS)			○	○		○	○			○							○																			○				○	
	Login Location (TS)			○				○			○							○																○								
	Login Time (TS)																																	○								
	Browsing Time (TS)	●	●															○								●			○			○		○								
	Browsing Navigational Information (TS)																○												○	○				○								
Transactional	Purchase Time (TS)	●	○	○	○	●	○	○	○		●	●	○	○	○		○	○	○		○	○	○		○	○	○	○	○	○	○	○	●	○	●		●				●
	Spending Amount (TS)	○	●		○	●	○		○		●		○	○	○	○	○		○		○	○	○		○			○	○	○	○	○	○		●		○				○
	Product Classification Information (TS)	●		○			○	●			○				○	●		○								○	●	○			○		○		●		○				○
	Purchase Equvivalent Categories (TS)						○																				●														
	Coupon Usage Information (TS)			○				○			○							●									○										○				
	Discount Amount (TS)	○		○							○							●					○											○							○
	Transportation Cost (TS)															●																									
Promotional	Promotional Transaction Information (TS)				○		○							○							●																				
	Promotional Status Information (TS)													○																											
	Effect of Promotion on Purchase				●																																				
Interactional	Review	○				○						●				○		●				○							○												
	Satisfaction Score/ NPS			●			○				○	●	○																												
	Complaint Information	○		●			○				○							●																							
Geo Demographic				○			○				○					○		○	○								○	○					○								○
Demographic		○		○	○		○		○	○	○	○		○				○	○					○			○	○				○		○	○	○	○				○
Behavioral		○	○	○	○		○	○			○	○					○	○						○		○			○	○		○		○			○				○
Transactional		○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○		○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○				○
Promotional					○		○							○							○																○				
Interactional		○		○			○	○			○	○	○			○		○				○							○												

Legend	
Used Feature	○
Top Feature	●

Subsequently, we present our observations regarding the specific features employed in the selected studies, with particular attention to those identified as most effective for churn prediction.

1. *Geo-demographic:* Features belonging to this category capture the demographic characteristics of the customer's location (e.g., population density, income levels, economic activity, neighborhood composition), rather than the customer's personal attributes. While indirect, these features can provide useful contextual signals for customer behavior.

2. *Demographic:* This category includes personal demographic details of the customer, such as gender, age, marital status, occupation, and education level. These features are typically available early in the customer lifecycle and can aid in early-stage churn prediction.

3. *Behavioral:* Behavioral features capture how customers interact with the business outside of transactions. In e-commerce settings, this may include browsing history, visit frequency, time spent on product pages, and preferred channels (e.g., mobile vs. desktop). These features provide insight into customer engagement and interest before purchase decisions.

4. *Transactional:* This group includes features derived from customer transactions, such as purchase time, amount spent, and item category. These features are used to create aggregated (or engineered) features like purchase frequency, total spend, average basket size, recency of the last purchase, and product categories purchased. These are among the most widely used and influential churn predictors.

5. *Promotional:* Promotional features refer to both the historical promotional activity of the retailer (e.g., discount campaigns, coupons) and the customer's responsiveness to these promotions. These features may include the number of promotional items purchased, redemption rates, and purchase patterns during sales events.

6. *Interactional:* Interactional data arises when customers voluntarily engage with the company beyond purchases. This may include customer reviews, feedback surveys, satisfaction scores, support tickets, and complaints. These features offer qualitative insight into the customer experience and sentiment.

As shown in Table 6, most studies incorporate transactional features, particularly Purchase Time (TS) and Spending Amount (TS). This widespread usage can be attributed to the general availability of such data, as most retailers, whether operating online or with loyalty programmes in physical stores, routinely collect these variables. This widespread use highlights their value in predicting churn. Many studies also use behavioral data, especially in e-commerce, where detailed web navigation logs complement transactional records, reflecting their predictive relevance. Customer profile attributes are also frequently used, indicating their potential value for churn modeling, though challenges in maintaining accurate and up-to-date profile data may impact its reliability in real-world applications.

The analysis of the selected studies revealed that top-ranked features are spread across various categories. This dispersion may result from different methodologies for evaluating feature importance, as varying techniques can yield different outcomes. The most informative features can also vary depending on the AI methods used, with more complex models potentially leveraging features more effectively. Additionally, feature engineering approaches influence how well features align with specific models, contributing to the variation observed in the ranking of top features.

4.2.3. RQ3: How have the available features been pre-processed and/or engineered before analyzing them using AI techniques?

Feature engineering plays a crucial role in machine learning, as it can significantly enhance both model performance and interpretability. Of the 41 selected studies, 34 reported the use of some form of feature engineering. Below, we summarize the techniques employed, as illustrated in Table 7.

Table 7: Classification of feature engineering techniques used in the reviewed studies.

Feature Engineering Method	**Study IDs**
RFM and its extensions	S3, S5, S6, S9, S8, S17, S21, S22, S28, S30, S31, S33, S35, S36, S37, S41
Feature selection	S2, S4, S9, S12, S18, S21, S19, S23, S27, S32, S35, S38, S40
Feature transformation/ aggregation	S5, S9, S11, S15, S16, S22, S23, S33, S35, S37
Customer segmentation techniques	S19, S24, S27, S29, S30, S37
Dimensionality reduction	S9, S16, S19, S22, S23

- *RFM and its extensions*: This category includes studies that use Recency, Frequency, and Monetary (RFM) analysis or its extensions to evaluate customer behavior and value. These derived features are often used to create informative intermediate representations that aid in churn prediction. RFM-based techniques are the most commonly used feature engineering approach in the reviewed studies.

- *Feature selection*: Studies in this category apply methods to identify the most relevant variables while eliminating redundant features. This process enhances model robustness and reliability, while reducing the computational cost of both training and inference.

- *Feature transformation and aggregation*: Studies in this category apply methods to modify, encode, or combine existing features to generate new, more informative representations. These transformations aim to create intermediate features that are more relevant for churn prediction than raw input data.

- *Customer segmentation techniques*: This category includes studies applying clustering or segmentation strategies to group customers based on shared behaviors or characteristics. Segment information provides additional context, potentially improving churn prediction accuracy.

- *Dimensionality reduction*: Studies that apply this method reduce the number of features by projecting data into a lower-dimensional space, while preserving essential

information. Dimensionality reduction facilitates more efficient learning and decreases computational complexity.

As shown in Table 7, RFM-based feature engineering techniques are the most commonly used, with 16 studies applying them, followed by feature selection methods (13 studies), feature transformation and aggregation (10 studies), customer segmentation (6 studies), and dimensionality reduction (5 studies).

4.2.4. RQ4: What AI techniques have been applied and found to be most effective in predicting customer churn in the retail sector?

The reviewed literature shows a broad application of AI techniques to address customer churn prediction in the retail sector. These methods span from Artificial Neural Networks (ANNs), Convolutional Neural Networks (CNNs), and Long Short-Term Memory networks (LSTMs), to more traditional stochastic methods, including Naive Bayes and Poisson Processes, along with various other approaches.

Table 8 summarizes the different AI techniques used by the reviewed studies. The symbols provide different information regarding the techniques' use: the empty circle define the method was used in the study (○); the filled circle indicate the best performing technique for the specific study (●); while the remaining symbol (■) is used only if the technique was the only one used in the study.

Among the tree-based models, Random Forest, Gradient Boosting, and Decision Trees are the most widely adopted and effective techniques. Their strong performance can be attributed to their capacity to model non-linear relationships and interactions between variables, as well as their robustness to outliers and noise in data. Support Vector Machines (for the distance-based classifiers) are also commonly employed, and are often highlighted for being the top-performing models, probably due to their generalization ability. ANNs are widely used across multiple studies. These architectures excel in capturing complex patterns in sequential or high-dimensional data, which is particularly present in churn prediction situations where customer behavior evolves. In addition to these dominant categories, several studies also explore linear models like Logistic Regression, which, despite being one of the most explored, is seldom ranked as the most performing model.

Considered together, there is no clear evidence that suggests that a dominating model outperforms all others. The effectiveness of an AI technique appears to depend on the characteristics of the dataset, the balance of the classes, and the quality of feature engineering. That being said, tree-based models and neural networks consistently emerge as strong contenders, showing their versatility in modeling customer churn in the retail domain. More recent and advanced AI techniques, such as Transformer-based models, and deep learning, have yet to be widely explored or proven effective in the context of churn prediction within the retail sector.

Table 8: AI techniques and their most effective applications across studies.

Category of AI techniques	AI Techniques	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	S17	S18	S19	S20	S21	S22	S23	S24	S25	S26	S27	S28	S29	S30	S31	S32	S33	S34	S35	S36	S37	S38	S39	S40	S41
Tree Based	Decision Trees	○		○	○		○	○	○	○								○						○			○			●									●	○	○
	Random Forest	○		●		○	○	○	○	○		●				●		○	○		●						○	○			○					○	●	●		○	
	ExtraTrees																	○																								
	Gradient Boosting Decision Tree		○						○																	●														○	●
	Light Gradient Boosting Machine								○															○																		
	eXtreme Gradient Boosting	○				○	○		○	○			●				●	○	●	○								●														
	CatBoost									○									●																							
	AdaBoost																	○				●																			○	
	Bagging Classifier								○									○																							●	
Distance Based	Support vector machines	●		○	●								○	○						○					●	○			●			●		●					○		
	K-Nearest Neighbors						○							○				○																								
	Multivariate Adaptive Regression Splines																																		●							
Linear and Discriminant	Ridge Regression /Classifer								○																○																	
	Linear Discriminant Analysis						○		○																		○									○				○		
	Quadratic Discriminant Analysis						○																				○															
	Logistic Regression		○			○	○	○	○	○		○	○				○		○	○		○			○		○	○				●		■		○	○	○	○		○	○
	Boosted Logistic Regression																										○															
Stochastic	Naive Bayes							○	○				○							○																				○	○	
	Possion Process																							●																		
Neural Networks	Artificial Neural Network		●	○	○	●		○	○		■			○	○				○	○	●		○			○		●		○		○		○			●	○				
	Recurrent Neural Network																						●																			
	Long Short-Term Memory																		●					○																		
	Convolutional Neural Network						○							●																												
	Restricted Boltzmann Machine						●																																			

Legend	
Used Model	○
Best Model	●
Only Model	■

4.2.5. RQ5: What are the most commonly used metrics to evaluate the performance of churn prediction techniques?

Each application of AI typically requires a distinct set of evaluation metrics to effectively demonstrate and compare the performance of different algorithms, pre-processing methods, and training strategies. This is because the objectives and success criteria of AI tasks vary significantly between domains. A metric that may be highly informative in one context could be less meaningful or even misleading in another. This variability is particularly evident in churn prediction, where several specific factors influence metric selection. Our review of the 41 selected studies highlights these specific challenges. Firstly, churn datasets are often highly imbalanced, with churners representing only a small fraction of the total customer base. As a result, evaluation metrics that account for class imbalance are essential. Secondly, in practical business scenarios, marketing interventions aimed at reducing churn come at a cost. Therefore, it is often more critical to avoid misclassifying non-churners as churners

(false positives) than to miss a few actual churners (false negatives). This cost-sensitive nature of churn prediction influences the choice and interpretation of performance metrics.

As shown in Table 9, the most frequently used evaluation metrics across the selected studies include Accuracy, Precision, Recall, F1/F scores, and AUC-ROC. These metrics are particularly well-suited for assessing the performance of churn prediction algorithms. For example, Recall indicates the proportion of actual churners correctly identified by the model, while Precision reflects the proportion of predicted churners who are indeed true churners. These metrics are directly linked to the practical implications of false negatives and false positives, as discussed earlier. The F1 and F scores offer a harmonic mean of Precision and Recall, summarizing the model's ability to balance these two aspects in a single value. In particular, the F score allows for adjustable weighting between Precision and Recall, providing flexibility to align the evaluation with specific business priorities. This makes it a valuable tool for deriving a single metric that captures both sensitivity and predictive accuracy. AUC-ROC is also widely used and regarded as a robust metric, especially in the context of class imbalance, which is common in churn prediction. It assesses the model's ability to discriminate between churners and non-churners across all classification thresholds. In addition to these five commonly used metrics, several studies employ more specialized or domain-specific measures. These bespoke metrics are typically chosen to reflect particular research goals or business considerations of the authors.

Table 9: Metrics to evaluate churn prediction techniques.

| | Study IDs | | |

environment, having different contexts, data types, and business sectors. These datasets are primarily obtained from:

- Open platforms such as Kaggle (e.g., the Retailrocket Recommender System Dataset D2 [71], D4 [72], D5 [73], D6 [74])
- Academic repositories (e.g., the Online Retail II dataset D3 [75], D9 [76])
- Corporate contributions (e.g., the Brazilian E-Commerce Public Dataset by Olist D1 [77], D7 [78], D8 [79])

As illustrated in Table 10, the datasets cover both online (E-Commerce) and offline (Brick-and-Mortar) scenarios and span multiple retail sectors, including merchandise, fashion, electronics, and food. These datasets differ not only in the retail sector they are related to, but also in structural characteristics, going from using only transactional records to multidimensional data including geo-demographic, behavioral, and interactional features.
It is interesting to note that the majority of the datasets are real-world datasets, but there is also a single synthetic dataset (i.e., the Retail Transaction Data) that is also used, showing attempts to simulate realistic churn behaviors for research purposes when actual data is limited or confidential.

This research question clearly uncovered the difficulties associated with the churn prediction task: each business, sector, dataset, and even study set a different definition for churn (as seen in Section 4.2.1). This inherent feature of this problem creates difficulty in comparing frameworks (needing to be equal in not only the dataset, but also the churn definition), making it impossible in this review to make any kind of comparison.

The datasets that already provide a churn flag, stating whether the customer has churned or not, show insufficient documentation, i.e., it is impossible to trace back what the adopted definition of churn is. This impacts reproducibility and points to the need for greater transparency and standardization in data reporting. This review encourages the creation of a clear dataset description and the use of practices that can help future researchers better assess, compare, and build robust churn prediction models in the retail domain.

Table 10: Available datasets.

ID	Dataset name	Pub. year	Retail sector	Synthetic/ Real	Online/ Physical	Features categories	Dataset size	Churn Flag	Churn Rate	Study IDs
D1	Brazilian e-commerce public dataset by Olist [77]	2018	fashion, mobile, electronics, appliance, etc.	Real	Online	Transactional	# Orders: 100k	F	NA	S3, S5, S10, S20
D2	E-commerce customer churn analysis and prediction [71]	2021	NA	NA	Online	Geodemographic, Demographic, Behavioural, Transactional, Interactional	# Rows: 5630	T	20.24%	S7
D3	Online Retail II [75]	2012	gift-ware	Real	Online	Transactional	# Instances: 1,067M	F	NA	S5, S8
D4	Retailrocket recommender system dataset [72]	2022	fashion, mobile, electronics, appliance, etc	Real	Online	Behavioural, Transactional	# Events: 2,7M # Unique Visitors: 1,4M # Product Information: 20M # Unique Items: 417k	F	NA	S26, S29
D5	Pakistan's largest e-commerce dataset [73]	2020/ 2021	fashion, mobile, electronics, appliance, etc.	Real	Online	Transactional	# Transactions: 500K	F	NA	S8
D6	Retail Transaction Data [74]	2017/ 2018	NA	Synthetic	NA	Transactional	# Transactions: 125K	F	NA	S25
D7	Carbo-loading [78]	Last update: 2023	Food	Real	Offline (Brick and Mortar)	NA	# Transactions: 5M+	F	NA	S24
D8	The complete journey [79]	NA	Food	Real	Offline (Brick and Mortar)	NA	# Transactions: 10M+	F	NA	S24
D9	A-feng dataset [76]	Last update: 2015	Food	Real	Offline (Brick and Mortar)	Geodemographic, Demographic, Transactional	# Transactions: 817K # Users: 32k # Items: 23,8K	NA	NA	S28

4.2.7. *RQ7: What are the challenges and limitations reported in using AI techniques for churn prediction?*

Fifteen of the 41 selected studies explicitly discuss challenges and limitations encountered in their research. These can be broadly categorized into five groups, as shown in Table 11. A detailed description of the categories is provided below.

- *Data availability:* Datasets used to experiment with methods for customer churn prediction are often limited in size, synthetic, or come from a single retailer or business sector. Incorporating data from multiple sectors and/or retailers could help build more resilient models and uncover deeper behavioral patterns.

- *Domain limitation:* This issue is closely linked to data availability and highlights the lack of data sources from diverse business domains. Much of the research on customer churn prediction is focused exclusively on either online or offline retail contexts, making it difficult to transfer findings or techniques across channels. Since customer dynamics can vary significantly across sectors, a domain-independent model needs to be trained on data with sufficient domain diversity to ensure generalizability.

- *Lack of comparative frameworks:* This issue emphasizes the absence of comprehensive testing and comparisons across a broad spectrum of AI frameworks and techniques. Without such comparisons, it is difficult to draw reliable conclusions regarding the relative performance of different approaches for churn prediction.

- *Computational resource limitation:* This issue pertains to constraints in computational resources available to researchers. Limited computational power prevents the exploration of optimal hyper-parameter configurations and restricts the ability to test larger or more complex models on substantial datasets, potentially limiting the performance and scalability of the proposed techniques.

- *Lack of robustness:* This issue concerns instances where technical inconsistencies arise, such as results that contradict the researchers' hypotheses or the presence of known weaknesses in the employed models that may introduce bias or skew results, undermining their theoretical validity.

Table 11: Challenges and limitations presented in the reviewed studies.

Challenges and Limitations	**Study IDs**
Data availability	S4, S16, S21, S28, S31, S36, S37, S38
Domain limitation	S4, S15, S29, S30, S31, S33, S37
Lack of comparative frameworks	S16, S22, S23, S30
Computational resource limitation	S15, S22
Lack of robustness	S9, S38

4.2.8. RQ8: What are the emerging trends and future directions for research in this area?

Of the 41 selected studies, 31 provided directions for future research. The proposed directions can be broadly grouped into seven categories which can be observed in Table 12. The details of the categories are as follows:

- *Exploring new frameworks:* Studies in this category suggest employing a broader range of AI techniques and fully exploiting these methods through hyperparameter tuning and architectural optimizations. The utilization of more advanced architectures, such as Recurrent Neural Networks, ensemble learning, transformers, and hybrid models combining supervised and unsupervised techniques, is expected to better capture sequential dependencies, rare events, and subtle patterns in customer behavior, potentially improving churn prediction performance. Additionally, to fully exploit the capabilities of AI models, studies propose optimization efforts, including hyperparameter tuning and dynamic adaptation to concept drift, which is gaining increasing attention. Techniques such as online learning, adaptive algorithms, and automated feature engineering pipelines are suggested to better handle evolving customer behaviors and seasonal trends.

- *Data and domain augmentation:* Studies in this category propose integrating richer and more diverse data sources, including social media sentiment, transactional history, weather patterns, spatial data, and socio-economic indicators. Together, these sources can offer a more holistic view of customer behavior and improve the generalizability of predictive models across various contexts. Furthermore, incorporating data from different retail sectors, such as fashion, electronics, and others, can enhance generalizability by ensuring that datasets capture a wider range of customer behaviors, which often vary between domains.

- *Customer segmentation and behavioral features:* Proposals in this category are similar to data augmentation efforts but focus specifically on incorporating behavioral features, such as web navigational data from e-commerce or hybrid retailers. Furthermore, some studies propose deriving intermediate features, such as customer segments, and providing this information to AI models to offer additional context about customers.

- *Temporal dynamics:* Studies in this category propose techniques aimed at more effectively capturing and leveraging the temporal dynamics of data. Such proposals include utilizing time series data directly, as well as exploring methods that allow models to better capture time-based patterns, such as experimenting with different window lengths and time series scales.

- *Explainability:* Studies propose techniques to improve the interpretability of AI models. This includes understanding the relative importance of various features by using tools such as Shapley values. Improving AI model explainability can foster greater trust in the models and provide deeper insights into the complex reasons behind customer churn.

- *ROI impact:* Studies emphasize considering the return on investment (ROI) implications when evaluating churn prediction models. For example, falsely predicting a non-churner as a churner has a higher negative financial impact than failing to identify a potential churner, as promotional resources would be wasted in the former case.

- *Real-world deployment:* Studies propose validating customer churn prediction models in real-world environments. Such validation would not only help assess churn prediction performance but also reveal the actual financial impact on businesses when these systems are integrated into promotional campaigns.

Table 12: Future directions presented in the reviewed studies.

Future directions	**Study IDs**
Exploring new frameworks	S3, S4, S5, S6, S7, S9, S10, S11, S13, S14, S18, S19, S20, S22, S23, S24, S25, S26, S28, S30, S33, S38, S41
Data & domain augmentation	S2, S3, S4, S7, S10, S11, S16, S19, S20, S21, S22, S26, S30, S31, S33, S36, S37
Customer segmentation and behavioral features	S6, S14, S15, S16, S21, S22, S28, S30, S33
Temporal dynamics	S4, S5, S15, S21, S23, S24, S31, S36
Explainability	S5, S7, S26, S41
ROI impact	S3, S12, S15, S37
Real-world deployment	S2, S14

5. Further discussion

This section expands upon the main findings of the systematic literature review by offering critical reflections and contextual analysis. The aim is to highlight cross-cutting insights and to examine how methodological and contextual differences influence the interpretation and application of churn prediction research.

5.1. Churn definitions

An analysis of the selected studies reveals a notable diversity in the definitions of customer churn. This variation suggests that churn definitions are often tailored to the specific characteristics of the retail context, including the type of retailer (e.g., e-commerce vs. brick-and-mortar) and the business sector in which they operate. Even among businesses of the same retail type, adopting distinct churn definitions may be appropriate, especially when customer behavior and distribution patterns differ across market segments. For instance, consider two grocery retailers: one located in a highly touristic area and the other in a more residential setting with stable purchasing patterns. In the former case, customer behavior is likely to be more variable and influenced by seasonality or transient visits, whereas in the latter, patterns may be more predictable and consistent. A fixed definition of churn (e.g., no purchases within a specific time window) may not perform equally well across both scenarios. Dynamic, customer-specific definitions that account for individual purchase history are likely to result in more accurate labeling and improved model performance in such contexts. Similarly, differences across retail sectors may also necessitate tailored churn definitions, as a single approach may not be equally suitable for all types of businesses.

5.2. AI techniques employed for churn prediction

While a variety of AI techniques have been applied to churn prediction, many studies continue to rely on conventional machine learning methods, such as decision trees, logistic regression, and random forests. The broader adoption of more modern architectures, such as transformer-based models or deep neural networks, could potentially alter the performance landscape. Given their ability to capture complex and nonlinear relationships between features, the relative importance of different variables in predicting churn could also change.

However, drawing definitive conclusions about the best-performing technique remains challenging due to several factors. First, the studies analyzed utilize different datasets, each with varying structures, feature sets, and levels of imbalance. Second, the definition of churn itself is not standardized and varies across studies: some define churn as complete inactivity, while others consider a decline in engagement within a specific timeframe. These inconsistencies hinder direct performance comparisons.

Although we grouped techniques into broad methodological clusters (e.g., neural networks, tree-based models), substantial variation exists within each cluster—ranging from model complexity and hyperparameter settings to architectural design. For instance, studies that apply artificial neural networks may differ widely in terms of the number of layers, node configurations, activation functions, and training regimes employed. As such, the results should be interpreted with caution, recognizing that similar-sounding methods may vary significantly in implementation and performance.

5.3. E-commerce vs Brick and Mortar Retailers

Research in customer churn prediction is more prevalent in the e-commerce domain, largely due to the availability of accessible and well-structured online datasets, including publicly available sources such as those hosted on Kaggle. E-commerce platforms benefit from extensive digital footprints left by users, which enable advanced predictive modeling using

sophisticated machine learning techniques, including deep neural networks and transformer-based architectures. The digital environment inherently supports continuous and fine-grained tracking of customer interactions, leading to more accurate and timely churn predictions.

In contrast, churn prediction in brick-and-mortar retail—particularly in sectors such as fast-moving consumer goods (FMCG) and grocery sectors—relies primarily on transactional data. The nature and granularity of available data in physical retail environments are more limited. While loyalty programs allow retailers to link purchases to individual customers, a portion of in-store transactions remains anonymous and cannot be attributed to specific customer profiles. As a result, the effectiveness of churn prediction in these settings depends heavily on the existence and adoption rate of robust loyalty systems.

In e-commerce, the requirement for user account creation allows for consistent tracking of individual customer behavior, extending beyond transactions to include web navigational behavior, such as pages visited, time spent on site, product views, browsing frequency, and revisit intervals. These non-transactional behavioral indicators offer early signals of churn intent, enabling proactive retention strategies.

By contrast, such behavioral data is largely inaccessible in physical retail. This reinforces the comparative richness of e-commerce data, which enables the integration of both transactional and behavioral signals in churn modeling. It is important to note, however, that retailers with both physical and online channels, often referred to as omnichannel retailers, can improve the accuracy of churn prediction by integrating data from both sources into their data warehouse. Interestingly, some of the studies that were focused on e-commerce did not utilize web navigational data, despite operating in a digital context where such features are readily available and potentially valuable.

Ultimately, churn prediction serves as a strategic tool to optimize marketing efforts, reduce customer attrition, and support revenue retention. Businesses—especially in data-rich environments like e-commerce—can closely align churn prediction with marketing strategies to enable timely and personalized retention interventions, maximizing the impact of promotional resources.

5.4. Final considerations

Synthesizing the results across our RQs provides a clearer picture of how AI is currently applied to churn prediction in non-contractual retail and why evidence accumulation remains difficult. First, most studies implement a largely standard predictive pipeline: churn is operationalized through study-specific labeling rules (Section 4.2.1), input signals are strengthened via engineered customer summaries (feature engineering is reported in 34 of 41 studies; Section 4.2.3) and performance is then benchmarked across a set of supervised learners where tree-based models and neural networks frequently emerge as strong candidates but no single approach dominates across contexts (Section 4.2.4). Second, the field's implicit "framework" is therefore predominantly methodological and data-driven rather than protocol-driven: model choices are strongly conditioned by dataset availability and by the adopted churn definition, which limits cross-study comparability. This is reinforced by the scarcity of reusable benchmarks (only 11 of 41 studies provide publicly available datasets; Section 4.2.6) and by the recurring single-source, context-specific settings emphasized among reported limitations (15 of 41 studies; Section 4.2.7), which together constrain domain generalizability across retail sectors and channels. Finally, the gap between technical results and

actionable retention decisions is visible in evaluation practice: the literature relies predominantly on standard classification metrics (Section 4.2.5), while future directions explicitly call for explainability, ROI-aware assessment, and real-world validation to connect predictive accuracy to intervention cost and business impact (Section 4.2.8). Taken together, these insights suggest that strengthening the paper's contribution is less about proposing yet another algorithmic variant and more about enabling transferable evidence through data/domain augmentation, clearer reporting of churn operationalization, and evaluation designs aligned with deployment and ROI constraints.

6. Threats to validity

The validity of our systematic review is subject to several interrelated threats across the different phases of the research process, including literature search, study selection, quality assessment, data extraction, and data synthesis. Following the guidelines of Petersen et al. [31], we organized potential threats into five dimensions: descriptive, theoretical, generalizability, interpretive, and repeatability. For every dimension, we identify the specific risks that could distort our findings, describe the concrete mitigation actions embedded in our protocol, and acknowledge the residual limitations that remain despite those safeguards.

Theoretical validity: Our evidence base may still miss relevant studies or misjudge their merit. File-drawer effects, language bias, and general publication bias can hide pertinent work, while venue reputation may inflate quality scores. Moreover, comparative bias can arise if the sequence in which papers are screened influences inclusion decisions. To mitigate these threats, we broadened the search string with synonyms using the PICOC framework and applied relaxed operators across multiple digital libraries, randomized the order of papers for each reviewer. Screening was performed in parallel, with disagreements resolved by consensus, and quality was assessed through explicit rubrics that we refined until full agreement.

Descriptive validity: A significant concern is whether the information we extracted from each primary study accurately reflects what the original authors reported. When compiling the data for our research questions, mapping extracted data to a common field could have introduced errors; for example, when tabulating features, the mapping was subject to researcher interpretation. We mitigated this risk through a shared, version-controlled data extraction form that tracks each mapping to the extracted original source, so that choices could be traced, discussed, and corrected in case of need.

Generalizability validity: While this review offers valuable insights into various aspects of customer churn prediction in non-contractual retail settings—including commonly used features, AI techniques, and evaluation metrics—the generalizability of some findings may be limited. The conclusions drawn from the analyzed studies are influenced by specific dataset characteristics, sector-specific dynamics, geographic context, and methodological choices such as feature engineering and model evaluation criteria. Therefore, claims regarding the effectiveness of particular techniques, the relevance of specific features, or the prevalence of certain metrics should be interpreted with caution when applied to different retail domains or operational contexts that are not represented in the reviewed literature.

Repeatability validity: For another team to replicate our study, the entire chain of evidence must be visible and reliable. We therefore provide details on the review protocol

design and execution (Section 3) and make the data extraction form along with the collected data publicly accessible.

Residual limitations: Despite the mitigation techniques adopted above, our search may still overlook some non-indexed studies, and quality scoring retains an element of human subjectivity. By publishing the full protocol and data extraction form, we keep these residual risks transparent and open to scrutiny.

7. Conclusions

This systematic literature review has thoroughly examined and synthesized the existing body of research regarding AI techniques applied to customer churn prediction within the retail sector, with particular attention to non-contractual environments characterized by gradual customer disengagement. The increasing digitalization of retail and higher competition make it easier for customers to defect and switch to competitors. Hence, understanding customer churn has become critical for businesses seeking strategic customer retention. Our investigation identified a notable increase in research interest from 2021 onwards, highlighting a timely expansion of academic attention likely driven by the growing demand for such research by the industry.

The wide range of research areas covered by the 41 analyzed studies reflects a clear interdisciplinary approach, driven by the complexity and multifaceted nature of churn prediction problems. In terms of churn definition, considerable variability was observed. The predominant approach adopted was based on a fixed time window of inactivity, followed closely by dynamic definitions. Each approach offers distinct advantages, from simplicity and ease of application to greater sensitivity in capturing subtle shifts in customer behavior. Nonetheless, the diversity of definitions highlights an ongoing challenge in the field: establishing standardized, universally applicable criteria for churn identification, which is fundamental for consistent comparison and evaluation of predictive models.

Regarding features for churn prediction, transactional features were found to be central, as they directly reflect customer interactions with retailers. Behavioral and demographic data were also widely employed, reiterating the importance of such data in capturing complex customer behaviors. Feature engineering techniques, particularly RFM analyses, were among the most effective and widely used methods, highlighting the importance of intelligent feature transformation to improve prediction performance.

The analysis of AI techniques revealed a clear preference for and effectiveness of tree-based algorithms, particularly Random Forest and XGBoost, as well as neural network approaches, including Long Short-Term Memory networks. More recent and advanced techniques, such as time-series analysis and Transformer-based models, have yet to be widely explored, despite their potential to capture complex patterns and temporal dependencies in customer behavior.

Model performance evaluation was predominantly conducted using Accuracy, Precision, Recall, F1-score, and AUC-ROC metrics, highlighting the importance of accurately identifying churners in scenarios where false positives and negatives bear significant operational costs.

However, several challenges persist. Notably, the limited availability of sufficiently large and diverse real-world datasets poses substantial constraints on model robustness and generalizability. Moreover, the computational cost and scalability of sophisticated AI models

remain significant barriers, necessitating optimized computational strategies and the adoption of scalable cloud-based infrastructure. When studies compare different AI techniques, the observed performance may not reflect each method's full potential, as resource constraints can limit the ability to optimize hyperparameters effectively for each method.

Emerging trends from this review clearly indicate opportunities for future research to expand the depth and breadth of churn prediction methodologies. Particularly promising areas include the integration of richer and more diverse data sources, such as social media data, detailed transaction histories, and socio-economic indicators, which can significantly enhance predictive accuracy and insights. Furthermore, advancing explainable AI models that leverage intrinsic attention mechanisms is strongly recommended to bridge the gap between model complexity and interpretability, enabling clearer strategic guidance for practitioners.

Additionally, the studies highlight key strategies for enhancing the predictive performance of churn models. These include the use of automated hyperparameter tuning to optimize model performance, the application of advanced ensemble methods, and the adoption of modern AI architectures such as transformer-based models and recurrent neural networks. Exploring spatial dynamics in churn behavior—especially the transition from partial to complete churn—also represents a compelling direction for future investigation.

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Appendix A. Review protocol details

Appendix A.1. PICOC and Search String

Table A.13: PICOC main terms and their synonyms.

View point	**Main term**	**Synonyms**
Population	Customers	Client, Consumer, Buyer, Purchaser, Acquirer
Intervention	Artificial Intelligence Techniques	AI, Machine Learning, ML, Deep Learning, DL, Intelligent System, Supervised Learning, Unsupervised Learning, Reinforcement Learning, Learning System, Automated Learning, Automated Reasoning, Computational Learning, Cognitive Computing, Machine Intelligence, Knowledge-based System, Intelligent Automation, Intelligent Agent, Smart Computing, Predictive Modeling, Ensemble Learning, Ensemble Model, Statistical Learning, Algorithmic Learning, Natural Language Processing, NLP, Computer Vision, Data Mining, Pattern Recognition, Optimization Algorithm, Genetic Algorithm, Evolutionary Algorithm, Metaheuristic Optimization, Automated Decision-making, Classification, Regression, Clustering, Time Series Analysis, Swarm Intelligence
Comparison	n.a.	-
Outcomes	Churn Prediction	(Churn, Attrition, Loss, Departure, Defection, Retention, Loyalty) AND (Prediction, Forecast, Detection)
Context	Retail Sector	Commerce, E-Commerce, Store, Shop, Point of Sale

Table A.14: Search strings for Scopus and Web of Science.

Digital Library	**Search string**
Scopus	*TITLE-ABS-KEY (("customer" OR "client" OR "consumer" OR "buyer" OR "purchaser" OR "acquirer") AND ("retail*" OR "commerce" OR "store" OR "point of sale" OR "shop") AND ("churn" OR "attrition" OR "loss" OR "departure" OR "defection" OR "retention" OR "loyalty") AND ("predict*" OR "forecast*" OR "detect*") AND ("artificial intelligence" OR "AI" OR "machine learning" OR "ML" OR "deep learning" OR "DL" OR "intelligent system" OR "cognitive computing" OR "machine intelligence" OR "knowledge-based system" OR "intelligent automation" OR "intelligent agent" OR "smart computing" OR "swarm intelligence" OR "intelligent agent" OR "predictive modeling" OR "statistical learning" OR "algorithmic learning" OR "supervised learning" OR "unsupervised learning" OR "reinforcement learning" OR "learning system" OR "automated learning" OR "ensemble learning" OR "ensemble model" OR "automated reasoning" OR "computational learning" OR "natural language processing" OR "NLP" OR "computer vision" OR "data mining" OR "pattern recognition" OR "optimization algorithm" OR "genetic algorithm" OR "evolutionary algorithm" OR "metaheuristic optimization" OR "automated decision-making" OR "classification" OR "regression" OR "clustering" OR "time series analysis"))*
Web of Science[7]	*SQ = ("customer" OR "client" OR "consumer" OR "buyer" OR "purchaser" OR "acquirer") AND ("retail*" OR "commerce" OR "store" OR "point of sale" OR "shop") AND ("churn" OR "attrition" OR "loss" OR "departure" OR "defection" OR "retention" OR "loyalty") AND ("predict*" OR "forecast*" OR "detect*") AND ("artificial intelligence" OR "AI" OR "machine learning" OR "ML" OR "deep learning" OR "DL" OR "intelligent system" OR "cognitive computing" OR "machine intelligence" OR "knowledge-based system" OR "intelligent automation" OR "intelligent agent" OR "smart computing" OR "swarm intelligence" OR "intelligent agent" OR "predictive modeling" OR "statistical learning" OR "algorithmic learning" OR "supervised learning" OR "unsupervised learning" OR "reinforcement learning" OR "learning system" OR "automated learning" OR "ensemble learning" OR "automated reasoning" OR "computational learning" OR "natural language processing" OR "NLP" OR "computer vision" OR "data mining" OR "pattern recognition" OR "optimization algorithm" OR "genetic algorithm" OR "evolutionary algorithm" OR "metaheuristic optimization" OR "automated decision-making" OR "classification" OR "regression" OR "clustering" OR "time series analysis")*

[7]The user interface of the Web of Science digital library was used to apply the defined search string SQ to Title, Abstract, and Keywords.

Table A.15: Search strings for IEEE and ACM.

Digital Library	Search string
ACM	*"query": Abstract:(("customer" OR "client" OR "consumer" OR "buyer" OR "purchaser" OR "acquirer") AND ("churn" OR "attrition" OR "loss" OR "departure" OR "defection" OR "retention" OR "loyalty") AND ("predict*" OR "forecast*" OR "detect*") AND ("artificial intelligence" OR "AI" OR "machine learning" OR "ML" OR "deep learning" OR "DL" OR "intelligent system" OR "cognitive computing" OR "machine intelligence" OR "knowledge-based system" OR "intelligent automation" OR "intelligent agent" OR "smart computing" OR "swarm intelligence" OR "intelligent agent" OR "predictive modeling" OR "statistical learning" OR "algorithmic learning" OR "supervised learning" OR "unsupervised learning" OR "reinforcement learning" OR "learning system" OR "automated learning" OR "ensemble learning" OR "automated reasoning" OR "computational learning" OR "natural language processing" OR "NLP" OR "computer vision" OR "data mining" OR "pattern recognition" OR "optimization algorithm" OR "genetic algorithm" OR "evolutionary algorithm" OR "metaheuristic optimization" OR "automated decision-making" OR "classification" OR "regression" OR "clustering" OR "time series analysis")) OR Title:(("customer" OR "client" OR "consumer" OR "buyer" OR "purchaser" OR "acquirer") AND ("churn" OR "attrition" OR "loss" OR "departure" OR "defection" OR "retention" OR "loyalty") AND ("predict*" OR "forecast*" OR "detect*") AND ("artificial intelligence" OR "AI" OR "machine learning" OR "ML" OR "deep learning" OR "DL" OR "intelligent system" OR "cognitive computing" OR "machine intelligence" OR "knowledge-based system" OR "intelligent automation" OR "intelligent agent" OR "smart computing" OR "swarm intelligence" OR "intelligent agent" OR "predictive modeling" OR "statistical learning" OR "algorithmic learning" OR "supervised learning" OR "unsupervised learning" OR "reinforcement learning" OR "learning system" OR "automated learning" OR "ensemble learning" OR "automated reasoning" OR "computational learning" OR "natural language processing" OR "NLP" OR "computer vision" OR "data mining" OR "pattern recognition" OR "optimization algorithm" OR "genetic algorithm" OR "evolutionary algorithm" OR "metaheuristic optimization" OR "automated decision-making" OR "classification" OR "regression" OR "clustering" OR "time series analysis")) OR Keyword:(("customer" OR "client" OR "consumer" OR "buyer" OR "purchaser" OR "acquirer") AND ("churn" OR "attrition" OR "loss" OR "departure" OR "defection" OR "retention" OR "loyalty") AND ("predict*" OR "forecast*" OR "detect*") AND ("artificial intelligence" OR "AI" OR "machine learning" OR "ML" OR "deep learning" OR "DL" OR "intelligent system" OR "cognitive computing" OR "machine intelligence" OR "knowledge-based system" OR "intelligent automation" OR "intelligent agent" OR "smart computing" OR "swarm intelligence" OR "intelligent agent" OR "predictive modeling" OR "statistical learning" OR "algorithmic learning" OR "supervised learning" OR "unsupervised learning" OR "reinforcement learning" OR "learning system" OR "automated learning" OR "ensemble learning" OR "automated reasoning" OR "computational learning" OR "natural language processing" OR "NLP" OR "computer vision" OR "data mining" OR "pattern recognition" OR "optimization algorithm" OR "genetic algorithm" OR "evolutionary algorithm" OR "metaheuristic optimization" OR "automated decision-making" OR "classification" OR "regression" OR "clustering" OR "time series analysis"))*
IEEE[8]	*("customer" OR "client" OR "consumer" OR "buyer" OR "purchaser" OR "acquirer") AND ("retail*" OR "commerce" OR "store" OR "point of sale" OR "shop") AND ("churn" OR "attrition" OR "loss" OR "departure" OR "defection" OR "retention" OR "loyalty") AND ("predict*" OR "forecast*" OR "detect*") AND ("artificial intelligence" OR "AI" OR "machine learning" OR "ML" OR "deep learning" OR "DL" OR "intelligent system" OR "cognitive computing" OR "machine intelligence" OR "knowledge-based system" OR "intelligent automation" OR "intelligent agent" OR "smart computing" OR "swarm intelligence" OR "intelligent agent" OR "predictive modeling" OR "statistical learning" OR "algorithmic learning" OR "supervised learning" OR "unsupervised learning" OR "reinforcement learning" OR "learning system" OR "automated learning" OR "ensemble learning" OR "ensemble model" OR "automated reasoning" OR "computational learning" OR "natural language processing" OR "NLP" OR "computer vision" OR "data mining" OR "pattern recognition" OR "optimization algorithm" OR "genetic algorithm" OR "evolutionary algorithm" OR "metaheuristic optimization" OR "automated decision-making" OR "classification" OR "regression" OR "clustering" OR "time series analysis")*

Appendix A.2. Inclusion and Exclusion Criteria

The EC/IC criteria of our study are as follows.

Exclusion Criteria.

(EC1) The study is a position paper or a short paper.

[8]Note that the IEEE Xplore digital library search query does not explicitly search for title, abstract, and keywords. Rather, it attempts to approximate the results by not placing any conditions on the search term. This has to be done due to the search limits imposed by the search engine.

(EC2) The study is not peer-reviewed.
(EC3) The study is a duplicate or an earlier version of another candidate paper.
(EC4) The full text is not available.
(EC5) The study is a secondary or tertiary study (Review, mini-review, survey, etc.)
(EC6) The study tests other authors' methodology (e.g., a study that tries to replicate the results of another study)

Inclusion Criteria.

(IC1) The study is written in English.
(IC2) The study focuses on Customer Churn Prediction in the Retail Sector using AI, based on customer and purchase data, which can be detailed as follows:
  (IC2.1) The study is strictly about Customer Churn prediction.
  (IC2.2) The domain strictly about retail?
  (IC2.3) Is the paper strictly using AI methodologies for churn prediction?
  (IC2.4) Is the churn prediction also based on data about customers' purchases (transactions) AND/OR customer profiling?